Limits & Derivatives

Section 1: Limits & Indeterminate Forms

1 Evaluate the limit: $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

a. 6

b. 3

c. 0

d. Does not exist

2 Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{4x^2 + 5}{2x^2 - 3x}$.

a. 2

b. 4

c. ∞

d. 0

3 Evaluate the trigonometric limit: $\lim_{x \to 0} \frac{\sin(5x)}{x}$.

a. 5

b. 1

c. 0

d. $\frac{1}{5}$

Section 2: Derivative Rules & Tangent Slopes

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4 If $f(x) = 4x^3 - 5x^2 + 2x - 7$, what is $f'(x)$?

a. $12x^2 - 10x + 2$

b. $12x^2 - 10x$

c. $4x^2 - 5x + 2$

d. $12x^3 - 10x^2 + 2$

5 What is the slope of the tangent line to the curve $y = x^2 - 3x + 4$ at the point where $x = 2$?

a. 1

b. 4

c. 2

d. -1

6 If $g(x) = (2x + 1)^4$, what is $g'(x)$?

a. $8(2x + 1)^3$

b. $4(2x + 1)^3$

c. $2(2x + 1)^3$

d. $8(2x + 1)^4$

Section 3: Critical Points & Extrema

7 For what value of x does the function $f(x) = x^2 - 6x + 8$ have a local minimum?

a. $x = 3$

b. $x = 6$

c. $x = -3$

d. $x = 4$

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8 What is the maximum value of $f(x) = -x^2 + 4x + 1$?

a. 5

b. 4

c. 2

d. 7

9 How many critical numbers does $f(x) = x^3 - 3x$ have on the real line?

a. 2

b. 1

c. 3

d. 0

Section 4: Higher Order Derivatives & Motion

10 If the position of an object is given by $s(t) = t^3 - 6t^2 + 9t$, what is its acceleration $a(t)$ at $t = 3$?

a. 6

b. 0

c. 12

d. -6

11 What is the second derivative $f''(x)$ of $f(x) = 2x^4 - 3x^2$?

a. $24x^2 - 6$

b. $8x^3 - 6x$

c. $24x^2 - 6x$

d. $12x^2 - 6$

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12 At what x-value does the curve $y = x^3 - 6x^2 + 5$ have an inflection point?

a. $x = 2$

b. $x = 4$

c. $x = 0$

d. $x = 6$

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